

ARTI308 - Lab 6 Summary

Linear Regression | Diabetes Dataset

Objective

Predict a quantitative measure of disease progression one year after baseline using 10 baseline predictors.

Work completed

1. Load: Read Diabetes.csv into pandas: 442 observations, 10 predictors, and one target. Data came from scikit-learn using scaled=False.
2. Explore: Used head(), info(), describe(), and correlation analysis.
3. Clean: Found no missing values, duplicates, or infinite values. All rows were retained.
4. Engineer features: Recoded sex categories 1 and 2 as a 0/1 indicator. Retained age, bmi, bp, and the six serum-related predictors. Excluded the target from X.
5. Prepare: Split the data into 265 training and 177 test observations using test_size=0.4 and random_state=101.
6. Train: Fitted a LinearRegression model on the training data, using the same model as the lab.
7. Evaluate: Predicted the test targets and calculated MAE, MSE, RMSE, and R². Compared with a training-mean baseline.

Test results

MAE	MSE	RMSE	R ²
46.1451	3286.9383	57.3318	0.4948

Conclusion

The model explained approximately 49.48% of test-set target variation. Its RMSE was 57.33, compared with 80.71 for the baseline. Prediction errors remain substantial. All notebook code cells ran without errors. R² is not classification accuracy.

Source: [scikit-learn Diabetes dataset documentation](#).